

JAHNAVI PRUDHIVI

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EDUCATION

Worcester Polytechnic Institute (WPI) | Worcester, MA

Expected May 2026

BS in Robotics Engineering & BS in Computer Science (GPA 3.80 / 4.00), Dean's List x5

- Coursework: Intro to Robotics, Robotics I Actuation, Robotics II Sensing, Robotics III Manipulation, Electrical Engineering, Control Engineering, Object-Oriented Programming, Embedded Computing, Machine Learning, Artificial Intelligence

SKILLS

Programming Languages: C++, Python, MATLAB, Java, C, C#, HTML/CSS/Typescript/Javascript, SQL, VBA, R

Frameworks and Development Tools: ROS2, PyTorch, MATLAB Simulink, AWS (S3, RDS, API Gateway), Git, OpenCV, OpenMV, React/Node/Next.js, Ubuntu, Jira

Design and Other: SolidWorks, OnShape, AutoCAD, Figma, Arduino, Raspberry Pi, Jetson

WORK EXPERIENCE

R&D Engineer and Researcher | Perception and Autonomous Robotics (Drone) | Worcester, MA

Aug 2025 - Present

- Developed deep-learning-based depth map models using a UNet architecture for lensless imaging, leveraging custom-fabricated lithographic phase mask optics and Intel RealSense data as ground truth
- Researched and applied lensless imaging techniques (Diffuser and Phase Masks) to put on a dragonfly-sized drone
- Implemented image reconstruction algorithms (ADMM, FISA, Nesterov) for image reconstruction and UNet for depth estimation for autonomy

R&D Engineer | The Autonomous Vehicle Mobility Institute (AVMI) | Worcester, MA

Aug 2024 - May 2025

- Used path planning methods in the simulation of autonomous and unmanned vehicles for exploration in severe terrain

API Support & Development Intern for SOLIDWORKS | Dassault Systèmes, Waltham, MA

May 2024 - Aug 2024

- Assisted and supported SOLIDWORKS and Connected CAD API for customers, partners, and internal teams
- Tested and automated sample code in VBA and C#, and controlled SOLIDWORKS functions
- Created project specifications based on customer and partner insights to develop new APIs for future SOLIDWORKS versions

AI Project Tester | ASSISTments | Worcester, MA

Feb 2023 - May 2023

- Tested image processing capabilities by integrating machine-learning techniques to strip PII from 100,000 image datasets

PROJECTS

ASL Gesture Recognition from Sign to Voice Output (Mudra AI) ([Link](#))

Nov 2025 - Nov 2025

- Built a real-time ASL recognition system with MediaPipe and OpenCV for hand and pose tracking for letter classification
- Trained a Logistic Regression model on 144 extracted landmark features, achieving 90% gesture recognition performance
- Integrated Google Gemini (NLP) for grammar correction and ElevenLabs (TTS) for speech synthesis voice output

Traffic Sign Recognition Using Deep Learning | WPI

Oct 2024 - Jan 2025

- Developed a custom CNN architecture using PyTorch and Google Colab to solve a multi-class classification problem with the German Traffic Sign Recognition dataset and fine-tuned pre-trained models, ResNet50 and VGG16, to improve model accuracy

4 Degree of Freedom (DOF) Arm Robot for Categorizing Objects | WPI

Aug 2024 - Oct 2024

- Programmed forward and inverse kinematics, differential kinematics, and trajectory planning to control an automated 3 DOF arm
- Developed ELP vision, camera calibration, and feedback control to identify and manipulate objects using MATLAB and Ubuntu

Software-Leader for SOLIDWORKS Robotics Program | Dassault Systèmes | Waltham, MA

May 2024 - Aug 2024

- Led the Software team of 10 interns to create a computer vision project using Python, ELP camera, and Raspberry PI 5 for April Tag detection to facilitate robot localization, and engineered a triangulation algorithm for positional accuracy and to locate objects
- Coded the Mecanum Drive train using Java and hosted lectures on coding, computer vision, and drive code methodologies

Escape Maze Robot | WPI

Mar 2024 - May 2024

- Built a four-robot autonomous system to line-follow white tape, climbing ramps using PID, perform tasks based on Open MV camera detect Apil Tags, using MQTT servers for communication with other robots, and wall-following to escape the maze

Autonomous Solar Panel Replacement Robot | WPI

Jan 2024 - Mar 2024

- Designed a four-bar mechanism in SolidWorks for a servo-actuated gripper and programmed an autonomous robot with a finite state machine (FSM) to remove and replace solar panels, integrating line-following algorithms, including limit switches, ultrasonic sensors, IR remote commands, and real-time feedback for efficient operation

LEADERSHIP

• **Secretary**, South Asian Student Association, WPI

Oct 2024 - Present

• **Software Leader (Robotics Project)**, Dassault Systèmes, Waltham

May 2024 - Aug 2024